

Does Syntactic Structure Make Language Models More Robust to Noise?

LSTM vs. RNN Subject-Verb Agreement Under Training-Data Corruption

Anonymous ACL submission

Abstract

1 Motivation

The core motivation of this study is to use controlled neural network experiments to investigate a long-standing debate in linguistics and cognitive science: how grammatical knowledge can be acquired from imperfect linguistic input.

According to Chomsky’s poverty-of-the-stimulus argument, children are exposed to noisy and incomplete language containing speech errors, disfluencies, and grammatical violations, yet still acquire complex hierarchical grammatical systems. This observation has often been taken as evidence that language acquisition depends on innate linguistic structure rather than solely on statistical learning from input. However, because it is neither ethically nor practically possible to systematically manipulate the linguistic environments of human children, the extent to which grammar acquisition depends on innate structure remains difficult to evaluate through direct experimentation.

Recent advances in computational linguistics provide a new opportunity to address this problem. Neural language models offer a uniquely controllable experimental platform in which both the learner and the learning environment can be manipulated. The inductive biases of different architectures can be interpreted as different forms of internal structure, while the quality of linguistic input can be controlled through the construction of training corpora. By systematically introducing grammatical corruption into the training data, we can simulate increasingly degraded linguistic environments and observe how different learners respond.

Beyond the broader question of whether grammatical knowledge can be acquired from noisy input, this study also seeks to understand how noise

affects the internal generalization strategies of language models. Natural language is fundamentally hierarchical, yet standard sequence models lack an explicit inductive bias for hierarchical structure. As grammatical evidence becomes increasingly unreliable, models may continue to rely on hierarchical syntactic representations, or they may gradually abandon them in favor of simpler linear heuristics based on surface word order. Tracking this transition provides a window into how grammatical noise reshapes the internal representations learned by neural language models.

Taken together, these considerations motivate three central questions. First, can language models acquire grammatical rules from systematically corrupted input? Second, does increasing grammatical noise cause models to shift from hierarchical generalizations toward simpler linear strategies? Third, does explicit structural inductive bias improve robustness to noisy linguistic environments? By answering these questions, we aim not only to characterize the robustness of neural language models, but also to contribute evidence to broader debates concerning grammar acquisition, inductive bias, and the learnability of hierarchical structure from imperfect input.

Condition	Rate	Acc. (all)	Attractor gap
baseline	0.000	0.961 ± 0.006	0.073 ± 0.023
low	0.004	0.958 ± 0.009	0.087 ± 0.030
medium-low	0.020	0.952 ± 0.014	0.103 ± 0.038
medium-high	0.100	0.934 ± 0.020	0.153 ± 0.051
high	0.500	0.708 ± 0.018	0.198 ± 0.033

Table 1: LSTM minimal-pair accuracy and attractor gap by training-noise rate (mean $\pm$ std over six runs). Accuracy is stable through 10% noise and collapses at 50%; the attractor gap grows monotonically throughout.

2 Background and Related Work

3 Methods

3.1 Data and Noise Injection

3.2 Leak-Free Evaluation Protocol

3.3 Minimal Pairs

3.4 Models

3.5 Metrics

4 Experimental Setup

5 Results

We evaluate every model on the same leak-free set of 19,819 minimal pairs (1,528 with an attractor, 18,291 without), held identical across all five noise conditions. Unless noted, reported values are the mean over six LSTM runs per condition (seeds {1, 2, 3}, run on both local MPS and Colab T4); $\pm$ denotes standard deviation across runs. The RNNG arm (Section 5.4) is a single full-scale seed evaluated on the same held-out set; additional seeds for error bars are in progress.

5.1 Noise degrades agreement, with a threshold near 50%

Overall minimal-pair accuracy is remarkably stable across the first four noise levels and then collapses at the highest rate (Table 1, Figure 1). Accuracy falls only 2.7 points as the error rate climbs from 0% to 10% ($0.961 \rightarrow 0.934$), but drops 22.6 points between 10% and 50% ($0.934 \rightarrow 0.708$). The model thus tolerates substantial agreement noise—up to roughly one corrupted verb in ten—before its grasp of the rule breaks down. This non-linear, threshold-like profile matches the first outcome anticipated in our proposal: input quality can be degraded considerably before grammar learning is affected, but beyond a critical point performance falls sharply toward chance.

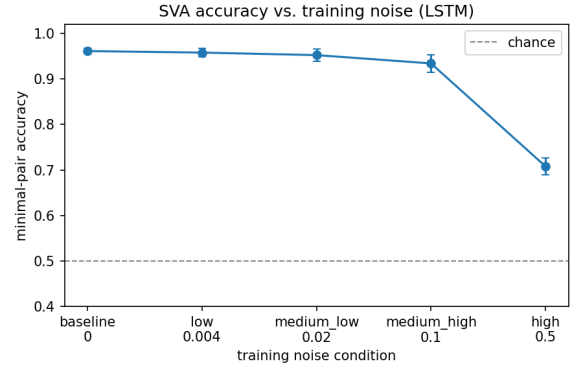


Figure 1: Overall minimal-pair accuracy vs. injected SVA noise rate. The dashed line marks chance (0.5). Performance is flat through 10% noise and falls sharply at 50%.

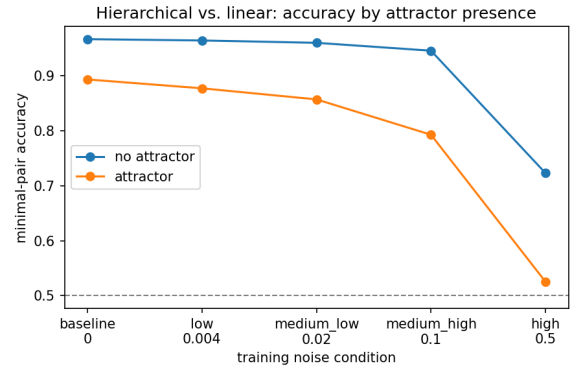


Figure 2: Accuracy on attractor vs. no-attractor items. The two cells diverge as noise rises: attractor accuracy approaches chance while no-attractor accuracy stays high until 50%.

5.2 Noise pushes the model toward the linear rule

Splitting accuracy by attractor presence isolates *which* rule the model relies on (Figure 2). No-attractor items—where the hierarchical and linear rules make the same prediction—stay above 0.94 until the 50% condition. Attractor items, where the two rules disagree, erode far faster: accuracy falls from 0.894 at baseline to 0.525 (near chance) at 50% noise. As a result the *attractor gap*—our index of reliance on the linear, nearest-noun heuristic—grows monotonically with noise, from 0.073 to 0.198 (Figure 3). Noisier training data drives the model away from the hierarchical rule and toward the linear shortcut, consistent with the “noise pushes models toward simpler generalization” branch of our second predicted outcome.

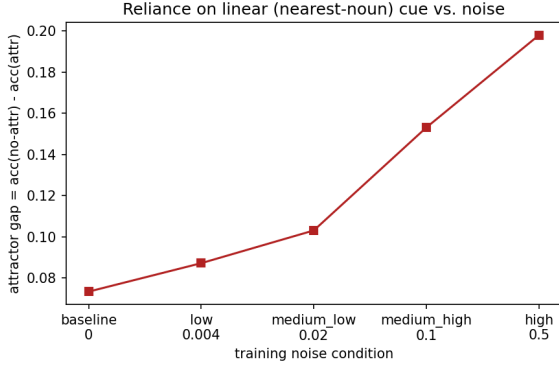


Figure 3: Attractor gap (no-attractor minus attractor accuracy) vs. noise rate. The monotonic rise indicates increasing reliance on the linear cue.

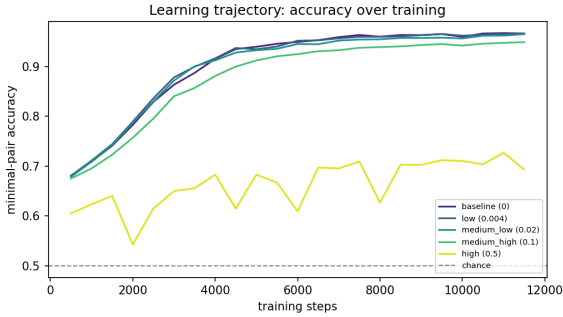


Figure 4: Minimal-pair accuracy over training steps, one curve per noise condition. Low-to-medium noise tracks the clean trajectory and converges (delay); the 50% condition plateaus well below the others (prevention).

5.3 Learning trajectories: delay vs. prevention

To ask *when*—not just whether—the rule is acquired, we checkpoint every 500 optimizer steps and re-score the full minimal-pair set at each point (Figure 4). The curves separate the two ways noise can hurt. For the baseline through medium-high conditions, accuracy follows nearly the same trajectory and converges to roughly the same endpoint (0.95–0.97): noise *delays* acquisition but does not prevent it. The 50% condition behaves qualitatively differently—it plateaus near 0.69 and never reaches the rule, i.e. *prevention*. The attractor gap reinforces this: for the clean model the gap shrinks over training as the hierarchical rule consolidates (0.148 $\rightarrow$ 0.054), whereas for the 50% model it *grows* with more training (0.143 $\rightarrow$ 0.197)—additional exposure to corrupted data entrenches the linear heuristic rather than correcting it.

5.4 Structure buys robustness: RNNG vs. LSTM

To test whether an explicit hierarchical bias changes how a model copes with noise, we ran the identical five-condition sweep with a Recurrent Neural Network Grammar (RNNG), which builds an explicit constituency structure as it processes each sentence. The RNNG is evaluated on the same held-out minimal-pair set as the LSTM (RNNG: 19,983 pairs; LSTM: 19,819), so overall accuracies are directly comparable. These RNNG values are a single full-scale seed (matched to the LSTM protocol at 150k sentences and eight epochs); seeds are still accruing, so we report point estimates without error bars and read them as a *direction* that the multi-seed run will tighten.

Two contrasts stand out (Table 2, Figures 5–6). First, the RNNG is more accurate than the LSTM at *every* noise level, and its margin widens exactly where noise bites hardest: the two architectures are nearly tied on clean data (0.979 vs. 0.961), but at 50% noise the RNNG holds 0.834 where the LSTM has fallen to 0.708—a 12.6-point gap. Across the full range the RNNG sheds only 14.5 accuracy points from baseline to 50% noise (0.979 $\rightarrow$ 0.834), against the LSTM’s 25.3-point collapse (0.961 $\rightarrow$ 0.708). The structural prior does not make the model immune—the threshold-like drop at 50% is still visible—but it substantially blunts it.

Second, and more tellingly, the RNNG barely drifts toward the linear heuristic. Its attractor gap stays low and essentially flat as noise rises, from 0.020 at baseline to 0.036 at 50% noise (a 1.8 $\times$ range), whereas the LSTM’s gap grows 2.7 $\times$, from 0.073 to 0.198 (Figure 6). The comparison is stark at the extremes: the RNNG’s gap under the heaviest noise (0.036) is only *half* the LSTM’s gap on perfectly clean data (0.073). Where the LSTM trades the hierarchical rule for the nearest-noun shortcut as its training data degrades, the RNNG keeps resolving agreement structurally almost regardless of noise. This is the signal our architectural hypothesis predicted: a built-in syntactic bias preserves the hierarchical generalization under input corruption that drives a sequential model to the linear one.

6 Discussion

When does the rule break? Two findings constrain when subject–verb agreement survives noisy training. First, the breakdown is *threshold-like*,

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		LSTM	RNNG	LSTM	RNNG
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low	0.004	0.958	0.979	0.087	0.030
medium-low	0.020	0.952	0.978	0.103	0.022
medium-high	0.100	0.934	0.971	0.153	0.034
high	0.500	0.708	0.834	0.198	0.036

Table 2: LSTM vs. RNNG minimal-pair accuracy and attractor gap by training-noise rate, on the same held-out set. LSTM values are the mean over six runs (Table 1); RNNG values are a single full-scale seed. The RNNG is more accurate at every noise level and keeps a near-flat, low attractor gap where the LSTM’s grows steeply.

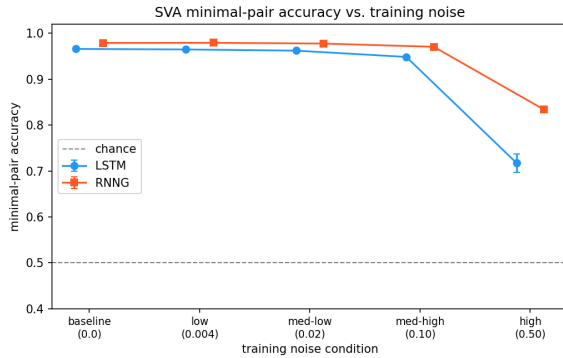


Figure 5: Overall minimal-pair accuracy vs. noise rate for the LSTM and RNNG. The RNNG sits above the LSTM throughout and separates sharply at 50% noise (RNNG single seed; LSTM mean of six).

not gradual: accuracy is nearly flat up to a 10% corruption rate and only collapses at 50% (Table 1). Small amounts of ungrammatical input are absorbed, much as a child’s grammar is not derailed by occasional speech errors. Second, the attractor gap is a more sensitive early-warning signal than overall accuracy: it climbs steadily (0.073 → 0.198) even while aggregate accuracy looks healthy, revealing that the model is quietly trading the hierarchical rule for the linear nearest-noun heuristic well before its overall performance gives way. The trajectory analysis sharpens the distinction between the two ways noise can act. Up to 10% noise, learning is merely *delayed*—every condition converges to the same accuracy given enough steps—whereas at 50% it is *prevented*: the model plateaus far short of the rule, and further training deepens rather than repairs its reliance on the linear cue.

Does structure buy robustness? Our central architectural question—whether a syntactically biased model (RNNG) withstands noise better than a

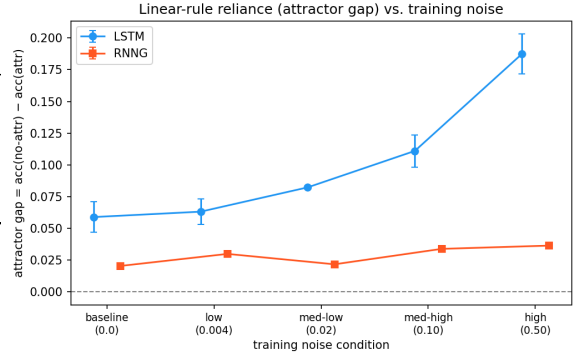


Figure 6: Attractor gap vs. noise rate. The LSTM’s reliance on the linear cue climbs steeply with noise (0.073 → 0.198); the RNNG’s stays low and nearly flat (0.020 → 0.036), below the LSTM’s clean-data gap even at 50% noise.

purely sequential one (LSTM)—the RNNG sweep now answers in the affirmative (Section 5.4). At matched noise the RNNG is more accurate than the LSTM throughout, and the advantage *grows* with noise, from under two points on clean data to 12.6 points at 50% corruption. The decisive contrast is in *which rule* each model keeps: the LSTM’s attractor gap climbs steeply as training data degrades (0.073 → 0.198), marking a drift toward the linear nearest-noun shortcut, whereas the RNNG’s gap stays low and almost flat (0.020 → 0.036)—under the heaviest noise it is still only half the LSTM’s clean-data gap. An explicit hierarchical prior thus preserves the structural generalization at noise levels where a sequential learner has already defaulted to the linear one, implicating architectural structure—not just input statistics—as a source of robustness. The finding is qualified: the RNNG still loses overall accuracy at 50% noise (0.979 → 0.834), so structure blunts rather than abolishes the threshold effect, and these RNNG values rest on a single (full-scale) seed pending the multi-seed replication. Read against the poverty-of-the-stimulus framing of Section 1, the result cuts toward the nativist side of the ledger: a structural bias measurably improves what is learnable from the same degraded input.

Cognitive-modeling angle. The pattern is suggestive for how structural knowledge might arise from statistical learning. The LSTM does not begin agreement-aware; the hierarchical rule is *acquired* over training, and in the clean setting the attractor gap shrinks as that rule consolidates. Noise does not flip a switch so much as shift the

balance between two competing generalizations—hierarchical and linear—both of which the data can support. Heavy enough noise makes the linear rule the better fit to the (corrupted) input, and the model follows the data. That a distributional learner recovers the hierarchical rule at all from this templatic corpus is itself evidence that the structure is more learnable from input than a strong nativist account would predict; that it abandons the rule under sufficient noise marks the limit of that learnability.

Threats to validity. Three caveats bound these claims; we treat them fully in Section 7. Our noise is synthetic and uniform—a fixed per-sentence flip probability—rather than the structured, frequency-dependent errors of natural input. The corpus is templatic and vocabulary-controlled, which aids control but limits external validity. Finally, the trajectory analysis is single-seed; the converging-versus-plateauing pattern is clear, but the precise crossover noise rate should be read as approximate until replicated across seeds.

7 Conclusion

Limitations

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A Reproducibility Details

B Additional Figures